

Geometric Spanning Trees that Minimizes the Wiener Index

Atishaya Maharjan

University of Manitoba
Geometric, Approximation, and Distributed Algorithms (GADA) lab

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This week

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- Think of suitable gadgets to prove the unbounded version is also NP-Hard.

Hardness Proof

Constant time approximation algorithm for hamiltonian path minimizing the wiener index

Definition (k-repairman problem (Contd))

Let $P_1, P_2, \dots, P_k$ be the paths assigned to the k repairmen, and each client is visited exactly once (by one path), then:

$$\min \frac{1}{|V|} \sum_{v \in V} t(v)$$

where $t(v)$ is the arrival time of the node v on the path it appears in.

Theorem (Theorem 2.5 [Dhamdhere et al., 2004])

Given a γ -approximation for the minimum k -repairmen problem on a metric d , we can obtain a 2γ -approximation for embedding the metric d into a k -spider in a non-contracting fashion to minimize the average distortion.

Constant time approximation algorithm for hamiltonian path minimizing the wiener index

- [Dhamdhere et al., 2004] showed that embedding a metric into a line (So the sum of pairwise distances in the line - The Wiener index of the path is minimized) can be approximated via the k -TRP.
- Moreover, [Sitters, 2021] gave a PTAS $(1 + \varepsilon)$ -approximation for the Euclidean k -TRP (and for planar graphs). Plugging $\gamma = 1 + \varepsilon$ provides a $2 + \mathcal{O}(\varepsilon)$ approximation factor.

Weak vs Strong NP-Hardness reductions

We begin with two related problems:

Definition (2-PARTITION)

Given n integers $A = \{a_1, a_2, \dots, a_n\}$, partition them into two groups A_1 and A_2 , where $A_1 \cup A_2 = A$ such that $\sum_{i \in A_1} i = \sum_{j \in A_2} j = \frac{1}{2} \sum_{a \in A} a = t$.

Definition (3-PARTITION)

Given n integers $A = \{a_1, a_2, \dots, a_n\}$ and no target sum, partition them into $\frac{n}{3}$ sets $A_1 \cup A_2 \cup \dots \cup A_{\frac{n}{3}} = A$ of equal sum: $\sum A_i = \frac{\sum A}{n/3} = t$. We will assume $3 \mid n$; otherwise the answer is no.

Weak vs Strong NP-Hardness reductions

- Weakly NP-hard: NP-hard “in the intended way”. Usually, we think of NP-hardness in terms of how hard your problem is, in terms of the encoding size (n) of your input. Thus, we allow numbers to have exponential value because even when a value is exponential, the encoding of the value is polynomial (Read in binary: $\log_2(\text{number} = 2^{n^c}) = n^c$, which is polynomial). Polynomial number of bits is in some sense a reasonable encoding.
- Strongly NP-hard: NP-hard even when problem is restricted to numbers of polynomial value (With respect to n , the number of numbers).

Key: **2-PARTITION** is Weakly NP-hard while **3-PARTITION** is Strongly NP-hard.

Definition (Euclidean Wiener Index Tree Problem (EWITP))

Given a set P of points in the Euclidean plane, a cost W , and a budget B , decide whether there exists a spanning tree T of P , such that $W(T) \leq W$ and $wt(T) \leq B$.

Theorem (Theorem 3 [Abu-Affash et al., 2023])

The EWITP is weakly NP-hard since they reduced it from 2-PARTITION.

Open problems from the geometric spanning tree paper

- Improve the running time of $\mathcal{O}(n^4)$ algorithm that constructs a spanning tree of minimum Wiener index when the points of P are in convex position.
- Is the problem of computing a spanning tree of minimum Wiener index for a set P of n points in general position NP-hard? The paper considered the variant where the total weight of the tree is bounded. If the total weight of the tree is unbounded, is it still NP-hard?

Next week tasks!

- Read and understand the hardness reduction and understand why bounding the weight of the tree is necessary.
- Think about either an efficient algorithm or a hardness reduction for the spanning tree minimizing the Wiener Index in general position.



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The end.
maharjaa@umanitoba.ca